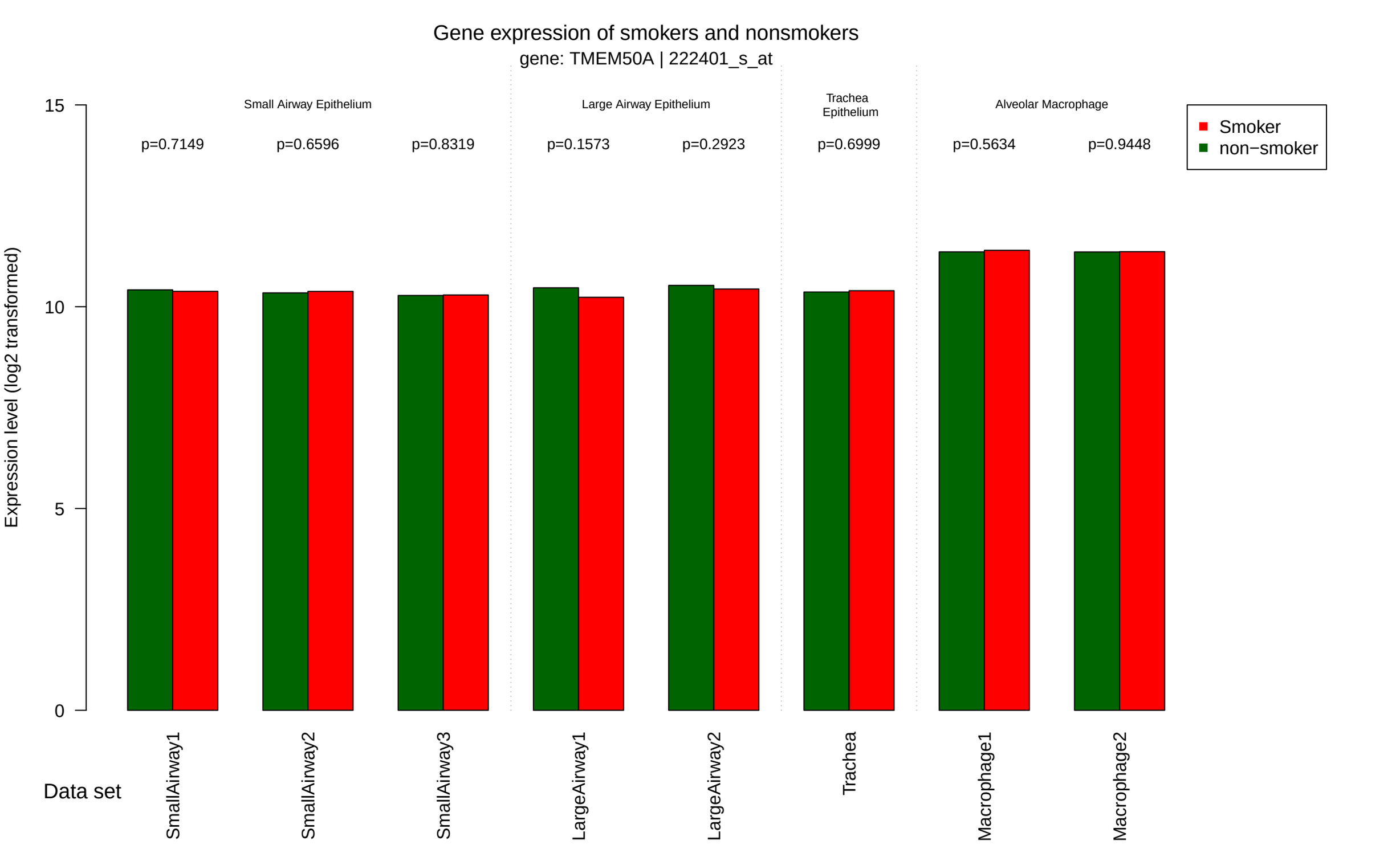
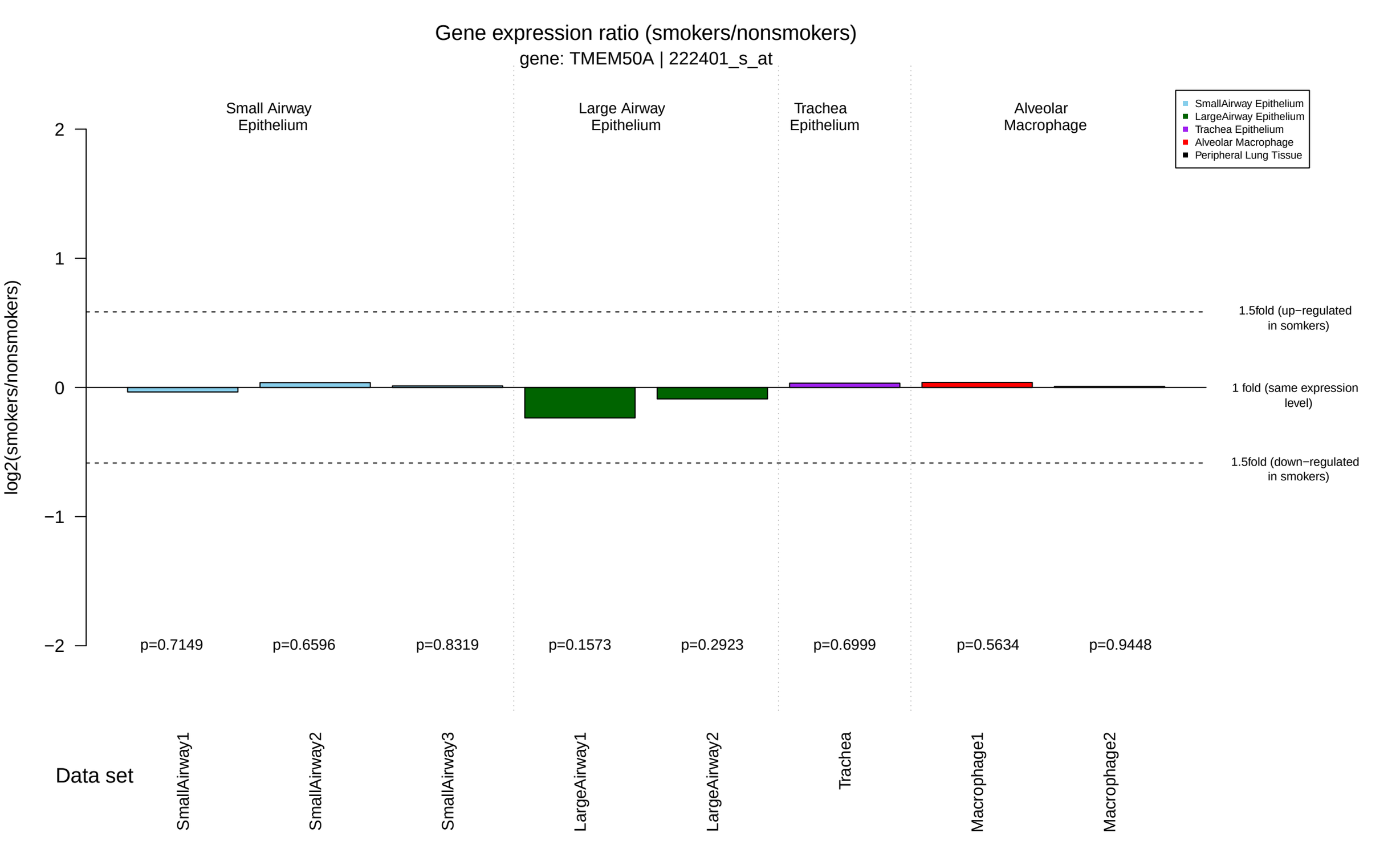
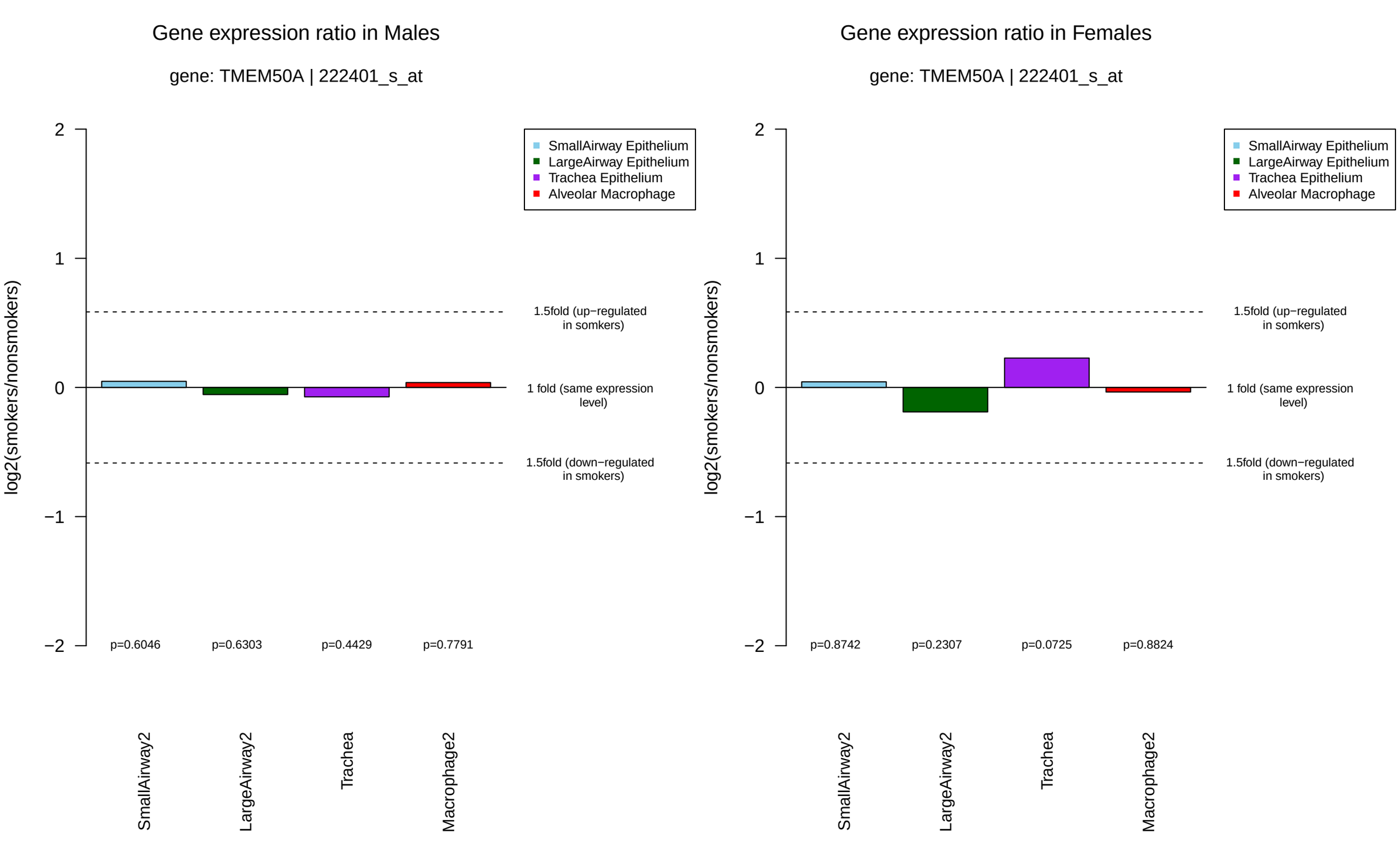
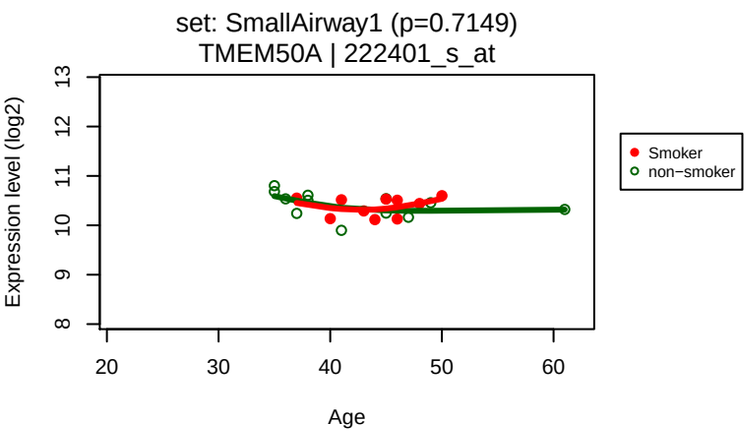
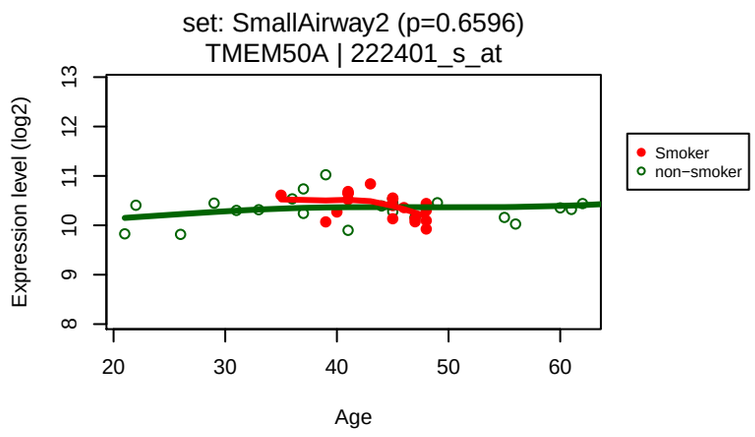
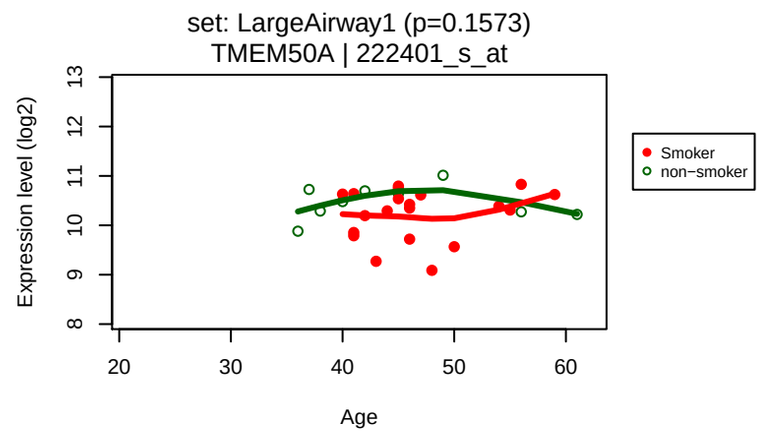

Gene expression of smokers and nonsmokers
gene: TMEM50A | 222401_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7149
p=0.6596
p=0.8319
p=0.1573
p=0.2923
p=0.6999
p=0.5634
p=0.9448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

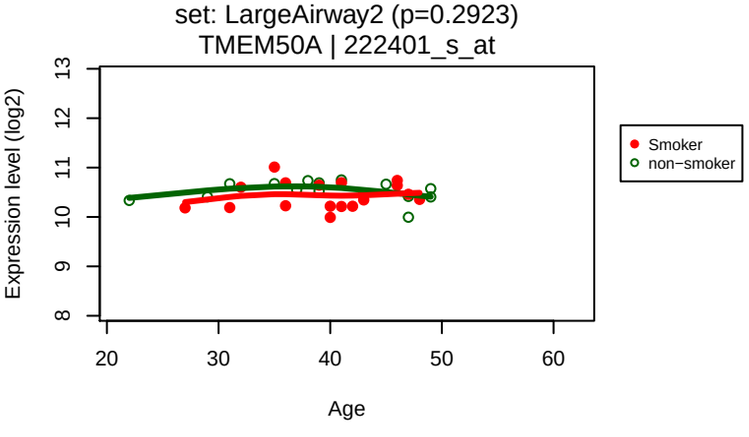

Gene expression ratio (smokers/nonsmokers)
gene: TMEM50A | 222401_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7149
p=0.6596
p=0.8319
p=0.1573
p=0.2923
p=0.6999
p=0.5634
p=0.9448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

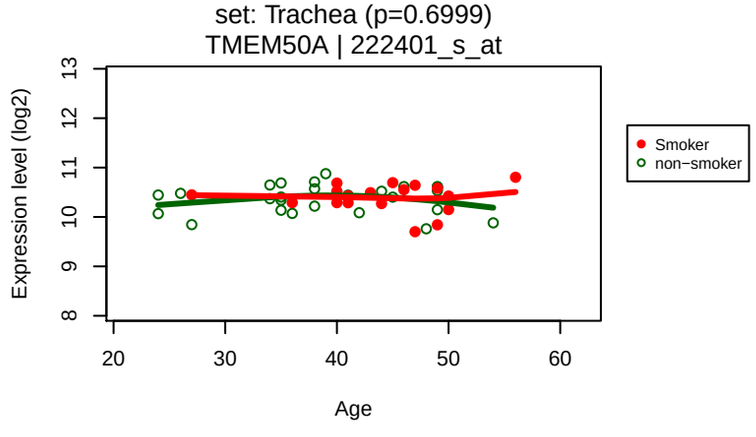

Gene expression ratio in Males
gene: TMEM50A | 222401_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6046
p=0.6303
p=0.4429
p=0.7791
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM50A | 222401_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8742
p=0.2307
p=0.0725
p=0.8824
SmallAirway2
LargeAirway2
Trachea
Macrophage2

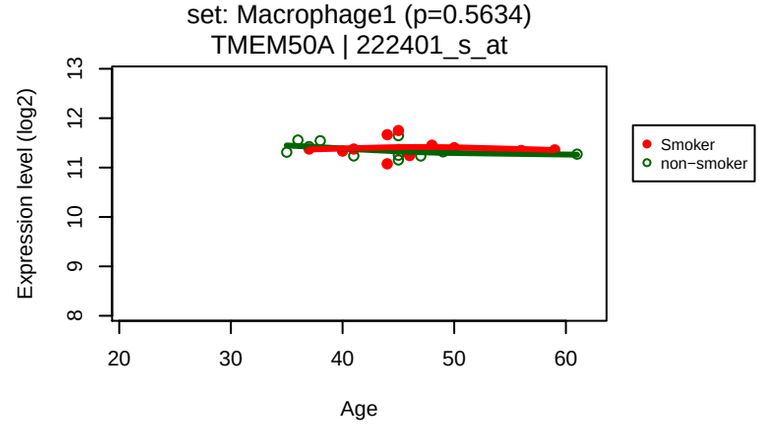

set: SmallAirway1 (p=0.7149)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker

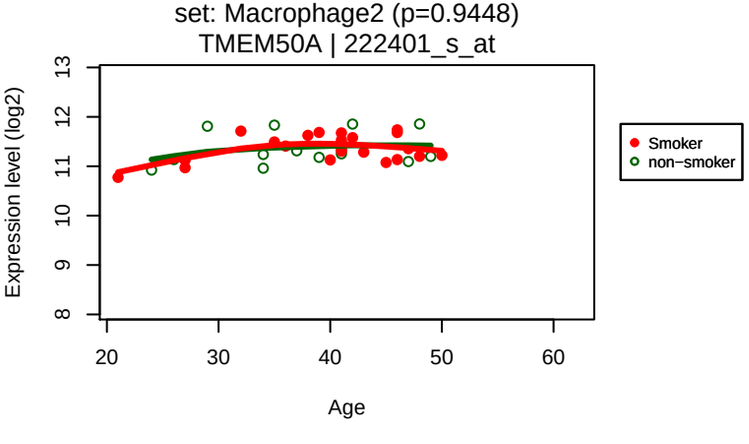

set: SmallAirway2 (p=0.6596)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker

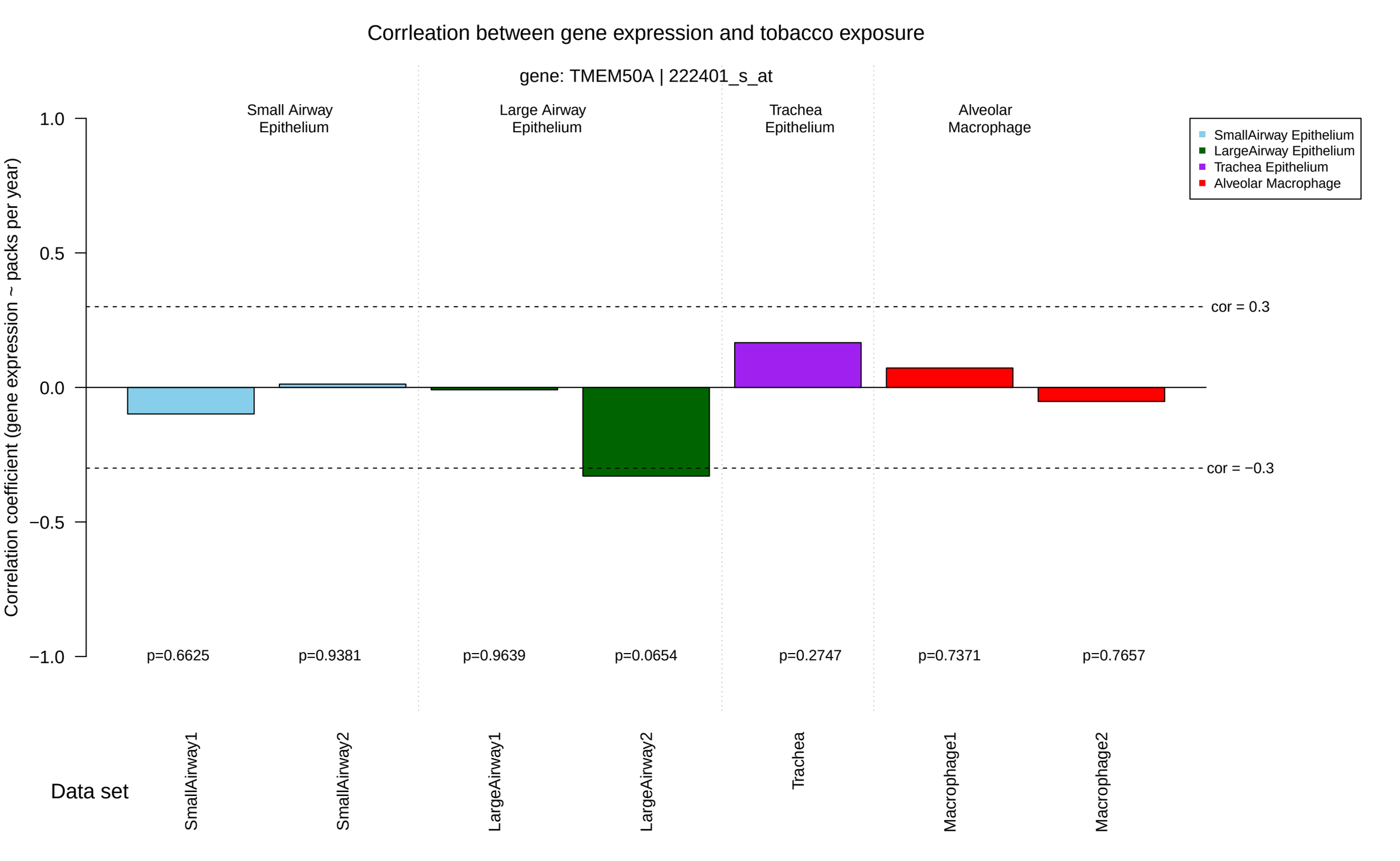

set: LargeAirway1 (p=0.1573)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2923)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6999)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5634)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9448)
TMEM50A | 222401_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM50A | 222401_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6625
p=0.9381
p=0.9639
p=0.0654
p=0.2747
p=0.7371
p=0.7657
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2